MODULE 22 · DOMAIN 2 - PROCESS

Building and controlling the schedule

Dependencies, the critical path, float, compressing and levelling, agile timeboxes, coordinating with operations, and analysing variation.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 22 of 49

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The project finishes when its longest chain of dependent work finishes.

Building a predictive schedule

01

Define activities

The actions needed to produce each work package.

›

02

Sequence

Which must come before which, and why.

›

03

Estimate

Durations, from module 21.

›

04

Develop

Calculate dates, the critical path and float.

›

05

Baseline

Approve it; measure progress against it.

ON THE EXAM

In PMBOK 8, these steps sit inside one process, Develop Schedule. The logic is unchanged.

Types of dependency

RELATIONSHIP	MEANING	EXAMPLE
Finish-to-start	B cannot start until A finishes	Pour concrete, then build walls. The most common
Start-to-start	B cannot start until A starts	Start testing once coding has started
Finish-to-finish	B cannot finish until A finishes	Documentation finishes when coding finishes
Start-to-finish	B cannot finish until A starts	Old system runs until the new one starts. Rare

ON THE EXAM

Leads let a successor start early; lags add waiting time, such as three days for concrete to cure.

Why a dependency exists

KIND	MEANING	CAN IT BE CHANGED?
Mandatory, or hard logic	Required by the nature of the work or a contract	No: you cannot roof before walls
Discretionary, or preferred	Chosen as best practice, not required	Yes, with some added risk
External	Depends on something outside the project	Not by the project alone
Internal	Between activities inside the project	Depends on its nature

ON THE EXAM

Fast tracking only works on discretionary dependencies. Mandatory logic cannot be overlapped.

A worked network

ACTIVITY	DAYS	AFTER	EARLY START	EARLY FINISH	LATE START	LATE FINISH	FLOAT
A	3	–	0	3	0	3	0
B	4	–	0	4	1	5	1
C	4	A	3	7	3	7	0
D	2	B	4	6	5	7	1
E	3	C, D	7	10	7	10	0

ON THE EXAM

Paths: A-C-E is 10 days, B-D-E is 9. The critical path is A-C-E, with zero float. B and D have 1 day.

Float, and what it tells you

- **Total float**: how long an activity can slip without delaying the project end date.
- **Free float**: how long it can slip without delaying the next activity's early start.
- The **critical path** has zero total float. Any delay there delays the project.
- Watch **near-critical** paths: a path with one day of float can become critical overnight.
- Negative float means the path is already later than a required date.

> ON THE EXAM
>
> A delay smaller than the activity's float does not move the end date. It does use up the float.

Crashing or fast tracking?

CRASHING

Add resources

- More people, overtime, faster equipment
- Increases cost
- Only on critical path activities
- Choose the lowest cost per day saved

FAST TRACKING

Overlap activities

- Do in parallel what was sequential
- Increases risk and possible rework
- Only on discretionary dependencies
- Usually no direct extra cost

THE TIEBREAKER

Money but no appetite for risk: crash. No money but some risk accepted: fast track.

Compression rules the exam uses

- Compress only the **critical path**. Shortening other work saves nothing.
- After each change, **recheck**: another path may have become critical.
- Crashing has limits: adding people to late work can slow it down at first.
- Consider options first: re-sequencing, reducing scope, or agreeing a new date.

> ON THE EXAM
>
> Speeding up non-critical activities is the classic wrong answer. It costs money and moves nothing.

Resource levelling or resource smoothing?

RESOURCE LEVELLING

Fix over-allocation

- Adjusts dates to match resource limits
- Can change the critical path
- May extend the end date
- Used when resources are the hard limit

RESOURCE SMOOTHING

Even out within float

- Adjusts only within available float
- Critical path unchanged
- End date unchanged
- May not remove every peak

THE TIEBREAKER

Levelling can move the end date. Smoothing never does, because it stays within float.

Critical chain, in brief

- Builds the schedule with **resource limits** in mind, not just logic.
- Removes safety from individual activities and pools it into **buffers**.
- A **project buffer** protects the end date; **feeding buffers** protect where other paths join.
- Progress is managed by watching how fast buffers are consumed.

> ON THE EXAM
>
> Critical chain pools uncertainty, rather than hiding it in every task. The same idea as honest estimates.

Schedules on agile and hybrid projects

- **Timeboxes**: fixed-length iterations; the team plans work to fit.
- **Release plans and roadmaps** show when capabilities are likely, forecast from velocity.
- **Milestones** link agile work to predictive work on hybrid projects.
- The outline's first schedule enabler: prepare the schedule **based on the selected development approach**.

> ON THE EXAM
>
> Agile schedules fix time and forecast scope. Hybrid schedules join the two at milestones.

Coordinating with other projects and operations

- The outline asks you to **coordinate with other projects and operations**.
- Shared people and equipment across projects: agree priorities early.
- Operational calendars: change freezes, peak seasons, maintenance windows, financial year end.
- A go-live planned inside an operational change freeze will not happen, however good the plan.

> ON THE EXAM
>
> The schedule lives inside the organisation's calendar. Check it before you commit dates.

Showing the schedule

FORMAT	BEST FOR
Bar chart, or Gantt chart	The team and managers: activities against time
Milestone chart	Sponsors and executives: key dates only
Network diagram	Analysing logic, dependencies and the critical path
Roadmap or release plan	Agile work: capabilities over time, as forecasts

ON THE EXAM

Match the format to the audience. Executives rarely need the full Gantt chart.

Controlling the schedule

- Compare actual progress with the **baseline**, regularly.
- For each slip, ask: **is it on the critical path?** Does it exceed the float?
- Update forecasts; take corrective action only where the end date or a milestone is at risk.
- Rebaselining is a change-control decision, not a way to make variance disappear.

> ON THE EXAM
>
> Analyse variation before reacting. A slip within float is information, not a crisis.

Project Lantern: the schedule

- **Critical path**: network upgrade, then kiosk installation, then migration rehearsal, then pilot go-live.
- **External dependency**: the regulator's approval, with a two-week lag before any clinic go-live.
- **Operations calendar**: no go-lives in December, the clinics' busiest month.
- The app's release plan meets the predictive schedule at one milestone per clinic.

> ON THE EXAM
>
> A predictive critical path, an external dependency, an operational constraint, and agile releases joined at milestones.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Durations and predecessors given	Find the longest path; float = late minus early
Late, money available, no extra risk wanted	Crash the critical path, cheapest first
Late, no money, some risk acceptable	Fast track discretionary dependencies
Over-allocated resource on the critical path	Levelling may extend the end date
Go-live clashes with an operational freeze	Coordinate with operations calendars in planning
An activity slips by less than its float	No end-date impact; monitor the reduced float

ON THE EXAM

Always ask: is it on the critical path, and how much float is there?

A network has five activities. A takes 4 days and B takes 6 days; both can start immediately. C (5 days) follows A. D (2 days) follows B. E (3 days) follows both C and D.

The project manager is assessing flexibility in the schedule.

How much total float does activity D have?

A 0 days

B 2 days

C 1 day

D 3 days

◆ PAUSE AND DECIDE

How much total float does activity D have?

A 0 days

D is not on the critical path. Only A, C and E have zero float.

B 2 days

D's own duration, not its float.

C 1 day

A-C-E = 12 days; B-D-E = 11. D can slip 1 day.

D 3 days

E's duration. Not relevant to D's float.

THE SIGNAL

Add up each path. Float on a non-critical path is the critical path length minus that path's length.

> A project is two weeks behind its contractual completion date. The sponsor has approved extra budget to recover, but insists that no additional risk of rework be introduced.
>
> The project manager has identified the critical path activities.

Which schedule compression approach should the project manager use?

A Fast track by overlapping critical path activities.

B Crash the non-critical activities with the lowest crash cost.

C Apply resource levelling across the schedule.

D Crash the critical path activities with the lowest cost per day saved.

◆ PAUSE AND DECIDE

Which schedule compression approach should the project manager use?

A Fast track.

Overlapping adds exactly the rework risk the sponsor ruled out.

B Crash non-critical work.

Spends money and moves the end date not at all.

C Resource levelling.

Resolves over-allocation; it can extend the schedule, not shorten it.

D Crash critical activities, cheapest first.

Uses the budget, adds no overlap risk, and targets the path that matters.

THE SIGNAL

Budget yes, extra risk no: crash, on the critical path, cheapest per day first.

> A software project is three weeks late. There is no budget for additional people or overtime.
>
> The team says it could begin building some components before the detailed design is fully signed off, accepting some risk of rework.

Which technique does this describe, and is it appropriate?

A. Fast tracking; appropriate, since no budget exists and some risk is acceptable.

B. Crashing; appropriate, since it shortens the critical path.

C. Resource smoothing; appropriate, since it keeps the end date.

D. Gold plating; inappropriate, since it adds scope.

◆ PAUSE AND DECIDE

Which technique does this describe, and is it appropriate?

A Fast tracking; appropriate.

Overlapping design and build; no extra cost; risk explicitly accepted.

B Crashing.

Crashing adds resources. There is no budget, and nothing is added.

C Resource smoothing.

Smoothing redistributes work within float; it does not recover lateness.

D Gold plating.

No scope is being added. The work is being overlapped.

THE SIGNAL

Starting the next step before the previous one finishes is fast tracking. No money, some risk: that fits.

A resource histogram shows the project's only database specialist allocated at 150 percent in weeks five and six. Both of her activities in those weeks are on the critical path.

No other specialist is available.

Which statement is most accurate?

A. Resource levelling will resolve the over-allocation without changing the end date.

B. Resource levelling is likely to extend the end date, because the affected activities are critical.

C. Resource smoothing will remove the over-allocation without any delay.

D. Crashing the activities will reduce her allocation.

◆ PAUSE AND DECIDE

Which statement is most accurate?

A Levelling, no end-date change.

Critical activities have no float. Delaying one delays the project.

B Levelling will likely extend the end date.

One of the critical activities must move later, and there is no float to absorb it.

C Smoothing, no delay.

Smoothing only uses float, and critical activities have none. It cannot help.

D Crashing reduces her allocation.

Crashing adds resources; with no other specialist, it cannot.

THE SIGNAL

Over-allocation on the critical path: levelling moves the end date, and smoothing has no float to use.

> A retail chain's new point-of-sale system was scheduled to go live in late November. Two weeks before, the operations director reminds the project manager that no system changes are permitted from November to early January.
>
> The freeze has been company policy for years.

What should the project manager have done?

A Crashed the schedule to go live before the freeze began.

B Asked the operations director for an exemption from the freeze.

C Checked operational calendars and constraints with operations while building the schedule.

D Proceeded with the go-live, since the project schedule was approved.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Crash to beat the freeze.

A reaction, not prevention, and the freeze may start before any recovery.

B Ask for an exemption.

A reaction, and freezes exist to protect peak trading.

C Coordinate with operations in planning.

The outline's enabler: coordinate with other projects and operations.

D Go ahead anyway.

Breaches a long-standing policy protecting the business at its busiest time.

THE SIGNAL

A long-standing operational constraint discovered late: it should have been in the schedule from the start.

> An activity to configure a reporting tool is running five days late. The schedule shows it has eight days of total float.
>
> The activity owner is worried and asks the project manager whether the project will be late.

What should the project manager do?

A Crash the activity immediately to recover the five days.

B Inform the sponsor that the project will finish five days late.

C Submit a change request to rebaseline the schedule.

D Confirm there is no impact on the end date, and monitor the activity now that its float has fallen to three days.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Crash it.

Spends money on an activity whose delay does not affect the end date.

B. Tell the sponsor it will be late.

Incorrect: a five-day slip within eight days of float does not move the end.

C. Rebaseline.

No baseline date is at risk. Nothing to rebaseline.

D. No end-date impact; monitor the three days left.

Correct analysis of variation, with attention on the shrinking float.

THE SIGNAL

Slip smaller than float: no end-date impact. But the float is now smaller, so watch it.

Find the longest chain. Protect it. Always ask about float.

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